

EX. NO: 4

IMPLEMENTATION OF VIGENERE CIPHER

AIM:

To implement the Vigenere Cipher substitution technique using C program.

ALGORITHM:

STEP-1: Arrange the alphabets in row and column of a 26*26 matrix.

STEP-2: Circulate the alphabets in each row to position left such that the first letter is attached to last.

STEP-3: Repeat this process for all 26 rows and construct the final key matrix. STEP-4: The keyword and the plain text is read from the user.

STEP-5: The characters in the keyword are repeated sequentially so as to match with that of the plain text.

STEP-6: Pick the first letter of the plain text and that of the keyword as the row indices and column indices respectively.

STEP-7: The junction character where these two meet forms the cipher character. STEP-8: Repeat the above steps to generate the entire cipher text.

PROGRAM:

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <string.h>
```

```
#include <ctype.h>
```

```
void encipher();
```

```
void decipher();
```

```
int main()
```

```
{
```

```
    int choice;
```

```
while (1)
{
    printf("\n1. Encrypt Text");
    printf("\n2. Decrypt Text");
    printf("\n3. Exit");

    printf("\n\nEnter Your Choice: ");
    scanf("%d", &choice);

    switch (choice)
    {
        case 1:
            encipher();
            break;

        case 2:
            decipher();
            break;

        case 3:
            exit(0);

        default:
            printf("\nInvalid Choice!\n");
    }
}

return 0;
}
```

```

void encipher()
{
    char input[50], key[50];
    int i, j;

    printf("\nEnter Plain Text: ");
    scanf("%s", input);

    printf("Enter Key: ");
    scanf("%s", key);

    printf("Encrypted Text: ");

    for (i = 0, j = 0; i < strlen(input); i++, j++)
    {
        if (j == strlen(key))
            j = 0;

        printf("%c", ((toupper(input[i]) - 'A' + toupper(key[j]) - 'A') % 26) + 'A');
    }

    printf("\n");
}

```

```

void decipher()
{
    char input[50], key[50];
    int i, j;

```

```
printf("\nEnter Cipher Text: ");
```

```
scanf("%s", input);
```

```
printf("Enter Key: ");
```

```
scanf("%s", key);
```

```
printf("Decrypted Text: ");
```

```
for (i = 0, j = 0; i < strlen(input); i++, j++)
```

```
{
```

```
    if (j == strlen(key))
```

```
        j = 0;
```

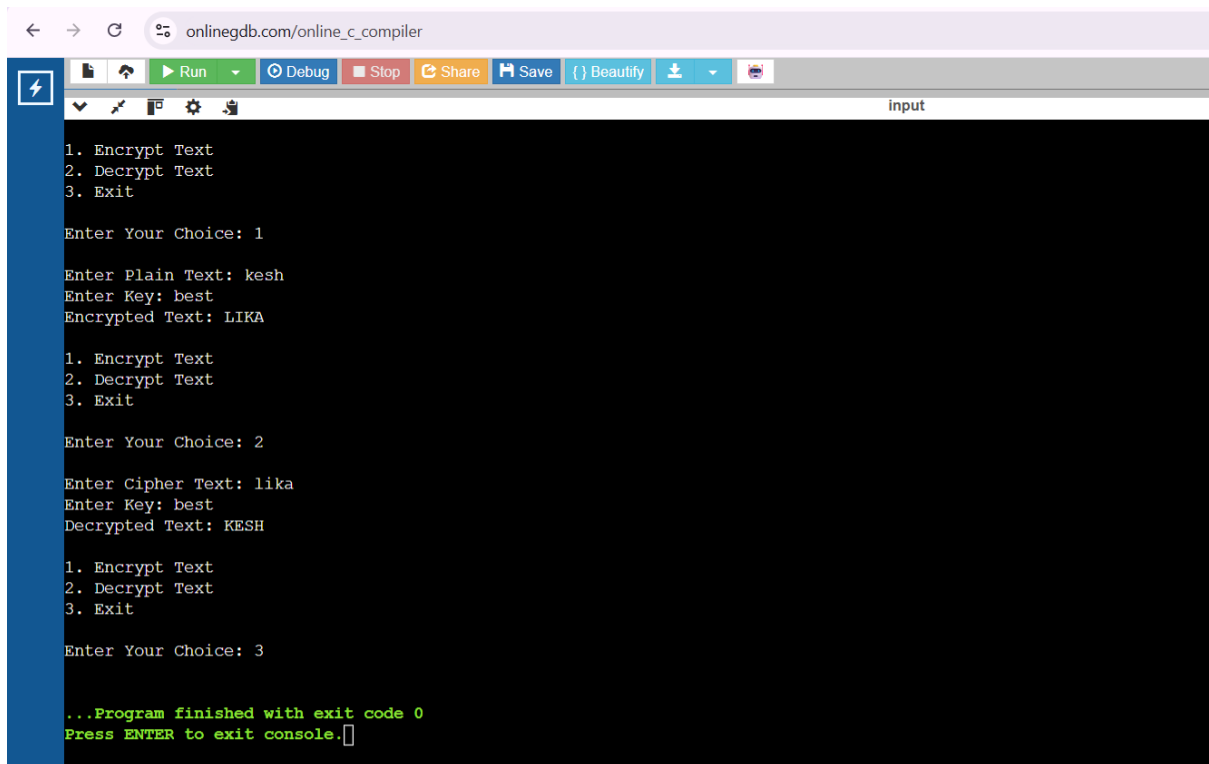
```
    printf("%c", ((toupper(input[i]) - 'A' - (toupper(key[j]) - 'A') + 26) % 26) + 'A');
```

```
}
```

```
printf("\n");
```

```
}
```

OUTPUT:



The screenshot shows a web browser window with the URL `onlinegdb.com/online_c_compiler`. The interface includes a toolbar with buttons for Run, Debug, Stop, Share, Save, Beautify, and a file icon. Below the toolbar is a dark-themed console window with the following text:

```
1. Encrypt Text
2. Decrypt Text
3. Exit

Enter Your Choice: 1

Enter Plain Text: kesh
Enter Key: best
Encrypted Text: LIKA

1. Encrypt Text
2. Decrypt Text
3. Exit

Enter Your Choice: 2

Enter Cipher Text: lika
Enter Key: best
Decrypted Text: KESH

1. Encrypt Text
2. Decrypt Text
3. Exit

Enter Your Choice: 3

...Program finished with exit code 0
Press ENTER to exit console.
```

RESULT:

Thus the Vigenere Cipher substitution technique had been implemented successfully.